

Sketch2Play: Runtime Generation of Playable 3D Game Content from Sketches

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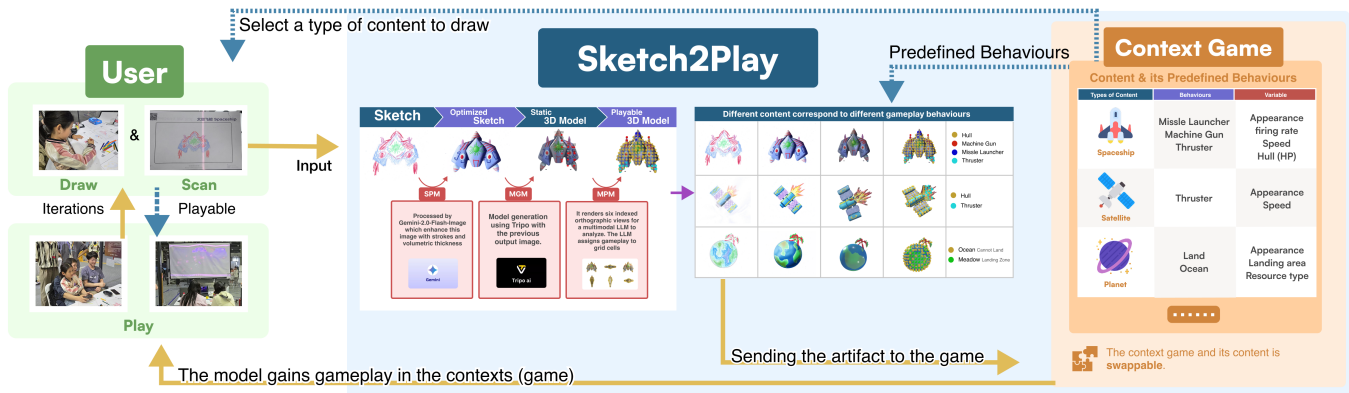


Figure 1: The Sketch2Play and companion game in action. Users draw a spaceship, scan it, and watch it transform into a playable 3D game content in real time. The system comprises three main modules SPM, MGM, and MPM and integrates with a companion game to bring the creation to life.

Abstract

Creating custom, playable 3D game content has traditionally required specialized skills and software, posing a high barrier for novices. This paper introduces Sketch2Play, a multistage game content creation system aimed at enabling users to make playable 3D game content using only hand-drawn sketches in runtime. User feedback indicates that Sketch2Play significantly lowers the barrier to creation, allowing children and non-professionals to easily turn their ideas into playable 3D content. This experience can greatly enhance users' creative agency, sense of ownership, and collaborative engagement. Sketch2Play demonstrates the immense potential of generative AI as a creativity accelerator and the new possibilities for user-centric interactive entertainment.

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CCS Concepts

• Human-centered computing → Collaborative and social computing.

Keywords

Human-AI co-creation, Procedural content generation, Playable media, Real-time interaction, User-generated content, Game prototyping

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INTRODUCTION

Children often cover their notebooks with adventurous thrusters exaggerated, cockpits glittering. However those drawings rarely pass beyond paper. The distance between an imaginative sketch and

a playable game object remains wide, because most titles rely on closed libraries and parametric reskins that grant only superficial customisation, and better customization requires creators to juggle professional tools, settings, and code. Research on user-generated content has highlighted the creative benefits of lowering production thresholds[2]. We therefore set out to create a system that would let even a complete novice, even a seven-year-old child, watch a freshly drawn spaceship take flight within moments. Our aim is to collapse the gap between freehand ideation and playable mechanics using generative AI, empowering and materializing everyone’s imagination and democratizing game development even further.

RELATED WORK

Efforts to democratise the creation of game content can be grouped into three principal lines. Easier game engines [1](e.g., RPGMaker) and asset marketplaces make it easy to make game or download prefabricated gameplay, yet still require additional skill. Parametric editors embedded inside games lower barriers further, but their templated controls impose tight limits on functional and aesthetic variety[2]. More recent studies show models and pipelines that translate text or images into detailed 3-D meshes[4] or scene[3] , yet they typically output static geometry that must still be rigged, scripted, and integrated before becoming playable. One study presented a solution using LLMs to make interactive world, but the model in it relies on existing asset library like Sketchfab[5].

SYSTEM ARCHITECTURE

The architecture of Sketch2Play is designed to be fully automated, requiring no manual intervention. A key architectural decision was to decouple content generation from gameplay logic. This enables Sketch2Play to operate across various contexts, generating playable 3D content according to different gameplay definitions. In this paper’s implementation, the context is provided by a companion game called Doodle Arena—a spacewar sandbox featuring spaceships, planets, and satellites, which supplies the specific behavior definitions.

The Sketch2Play process unfolds in three modules: SPM, MGM, and MPM. Each module builds upon the previous, progressively transforming the sketch into a fully functional game artifact ready for deployment in a target game environment.

The sketch processing module (SPM) begins when a user places a drawing in the scanner and hits the scan button. Once SPM detects a new image, it crops away instructional text and marginal notes, leaving only the sketch. The cleaned image is then processed by Gemini-2.0-Flash-Image, a multimodal LLM capable of editing images, which enhances the sketch with strokes and volumetric thickness, making them more suitable for 3D reconstruction.

The model generation module (MGM) consumes the enriched image and uses Tripo2.5, a generative 3D model that produces detailed geometry and textures from text and images. This enables the generation of a 3D mesh that preserves the original sketch’s silhouette and color characteristics.

The model processing module (MPM) transforms the static mesh into playable content. It renders six orthographic images of the mesh and overlays each with a square matrix whose cells carry unique indices. These indexed views provide a coarse spatial

reference frame for multimodal LLM reasoning. To avoid LLMs’ tendency to repeat structural tokens, the system restructured the output format into index-label pairs. The 3D model, along with a predefined behaviour definitions from the companion game, is provided to the multimodal LLM, which assigns behaviours such as engines, missiles, or turrets. Behaviour parameters like thrust, firing rate, and HP scale with the number of assigned cells. A filter removes “orphan” indices disconnected from neighbouring cells, ensuring all gameplay triggers remain grounded on the mesh’s surface.

Finally, the system presents the model in GLB format, accompanied by a JSON manifest, to the context environment— in this case, the Doodle Arena game—via a local webhook. The game reads the URL, fetches the model and manifest, and automatically attaches the appropriate MonoBehaviours based on the manifest, granting the 3D model gameplay in runtime, with no recompilation required.

In the context of Doodle Area game, in addition to spaceships, we also provide gameplay behaviour for planets and satellites to increase gameplay variety.

DEMONSTRATION ENVIRONMENT

The public demonstration consists of a laptop, a scanner, a set of 24-colour crayons and papers with text instructions. Users draw a spaceship within the paper, put the drawing in the scanner, and press the scan button. A progress bar displays the status of the three pipeline stages, providing a glimpse into system internals. Upon completion, the spaceship spawns in Doodle Area game. The player receives a Joy-Con gamepad to steer, accelerate, and deploy projectiles. A second player can be activated by scanning another sketch. A separate projector displays real-time video of the progress and game scenes to spectators, fostering social engagement and co-creation.

PRELIMINARY EVALUATION

We conducted a hands-on session with two groups: twelve families and multiple adults. The session began with a brief tutorial, before they entered a thirty-minute free exploration phase.

Deployments revealed that players rapidly internalised the relationship between drawing and in-game behaviour. One child iteratively enlarged thrusters after observing insufficient acceleration. Parents frequently engaged in descriptive scaffolding, such as suggesting that the addition of an engine can make the spaceship move faster. Among the 14 children participants, 12 clearly expressed emotional excitement when their drawing was transformed into a playable object. Several used phrases such as “That’s my spaceship!” and “It looks just like I drew it!”, indicating a strong sense of ownership and recognition.

Some professional users who tested the system expressed enthusiasm about integrating Sketch2Play into creative workflows, especially for rapid prototyping and early-stage ideation. They also noted concerns regarding generative AI’s unpredictabilities.

FUTURE WORK

“Sketch2Play” currently demonstrates a feasible and real-time system for transforming hand-drawn sketches into playable 3D content

through AI assistance. We see potential both technologically and practically:

Expanded evaluation. Current insights are primarily based on observational data. We plan to conduct controlled user studies with a larger and more diverse participant pool to more rigorously assess usability and creative impact.

New contexts. While our current implementation is built around a space shooter gameplay context, the Sketch2Play pipeline is designed to be modular and adaptable. This flexibility opens the door to diverse applications. Furthermore, Sketch2Play can serve as a rapid prototyping tool for professional designers, dramatically reducing iteration time in the form of Unity Editor plugin.

CONCLUSION

Sketch2Play demonstrates that a carefully orchestrated pipeline of computer vision, generative modelling, and semantic interaction mapping can transform informal, expressive sketches into coherent, rich, and playable content within minutes. While the system incorporates techniques to interpret hand-drawn input, it still faces inherent limitations when dealing with highly abstract or idiosyncratic sketches. Nevertheless, Sketch2Play dissolves the boundary between author and audience, prompting a rethinking of how games might be conceived. This suggests that the future

of digital interaction may be less about navigating pre-fabricated worlds and more about drawing new ones into existence.

Acknowledgments

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